

Facial Image Colorization Using Pix2Pix: Bringing Shades to Life

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Abstract

From the restoration of historical photographs to the generation of creative content, the ability to manipulate and transform images in semantically coherent ways has significantly advanced due to conditional generative adversarial networks (cGANs). These models effectively bridge the gap between human perception and computational visual understanding. This work applies the Pix2Pix architecture for automatic colorization of grayscale images, focusing on the CelebA dataset of human faces. The proposed model employs a U-Net-based generator with skip connections coupled with a PatchGAN discriminator. Training was optimized using techniques such as label smoothing, label flipping, and progressive learning rate decay to enhance stability and convergence. Performance evaluation was conducted using quantitative metrics, including Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), DeltaE, and chromatic accuracy, by systematically comparing generated images to ground truth. Results demonstrate that the model produces realistic and perceptually coherent colorizations, achieving performance comparable to or exceeding established baselines. These findings confirm the suitability of the Pix2Pix architecture for image colorization tasks and suggest promising applications in digital restoration, creative media production, and facial image enhancement.

1. Introduction

In recent years, the development of deep learning architectures has enabled the emergence of numerous approaches to automatic colorization, with a particular focus on human faces. Among these, Pix2Pix [1] has established itself as one of the most effective models for supervised image-to-image translation. The architecture relies on a conditional generative adversarial network (cGAN) composed of two neural networks—a generator and a discriminator—that are trained in opposition to refine the quality of the generated outputs. Pix2Pix learns a direct mapping from grayscale inputs to colored outputs, maintaining both global consistency

and local detail. This makes it especially well-suited to tasks like face colorization, where spatial structure and fine features must be preserved during the transformation process.

Compared to earlier approaches such as that of Zhang et al. [2], which frame colorization as a classification task in the *ab* chromatic channels (Lab color space), Pix2Pix produces visually richer results, avoiding the desaturation typically observed in purely discriminative methods. More recent methods like FCNet [3] incorporate facial semantic segmentation to guide the colorization process but require detailed annotations and more complex preprocessing pipelines. Advanced architectures such as ColTran [4] achieve photorealistic results but at the cost of higher computational requirements and structural complexity, making them less accessible for medium-sized datasets and straightforward supervised tasks.

Pix2Pix, on the other hand, offers a balanced solution in terms of visual quality, implementation simplicity, and training stability. These characteristics make it particularly well-suited for automatic face colorization in supervised scenarios, where a direct correspondence between grayscale and color reference images is available.

This work proposes an optimized implementation of the Pix2Pix architecture for facial image colorization, based on a U-Net generator with skip connections to preserve local structure and a PatchGAN discriminator that evaluates the realism of local image patches. The training process was stabilized using well-established techniques in GAN literature, including label smoothing, label flipping, progressive learning rate decay, and asynchronous discriminator updates.

The system was evaluated on a reduced version of the CelebA dataset, operating in the Lab color space and employing both objective metrics (PSNR, SSIM) and perceptual measures (DeltaE, Chroma Accuracy). The results demonstrate that the model produces realistic, structurally faithful, and chromatically coherent colorizations consistent with the target facial domain.

2. Data

The experiments used a reduced version of the CelebA (CelebFaces Attributes) dataset¹, consisting of human face images.

It is a large-scale collection of celebrity face images widely used for face attribute analysis, recognition, and generation tasks. For this work, it was used a subset of 25,000 images selected from the original dataset of over 200,000 images. The selection maintained diversity across facial characteristics including pose, expression, age, and ethnicity while reducing computational requirements. The original dataset was published by Liu *et al.* [5] and is available under non-commercial research licenses. The CelebA dataset was selected due to its:

- High-quality face-centered images with consistent framing;
- Diverse representation of human faces
- Standardized alignment (aligned version);
- Established benchmark status in GAN-based image translation tasks.

2.1. Dataset Partitioning

The 25,000 images were divided into three distinct sets to ensure proper evaluation:

- **Training set:** 20,000 images (80%) used for model parameter learning;
- **Validation set:** 2,500 images (10%) used for hyperparameter tuning and early stopping;
- **Test set:** 2,500 images (10%) used exclusively for final performance evaluation.

The dataset was organized into a directory structure with separate folders for training, validation, and test sets. This folder-based organization simplified data loading and ensured strict separation between the different data splits, preventing any information leakage between training and evaluation. The partitioning was performed randomly while preserving the overall distribution of attributes across the three sets.

2.2. Hyperparameter

To reduce computational requirements during hyperparameter optimization, a smaller subset of 2,000 images (10% of the training set) was used for initial experimentation. This approach allowed us to:

- Rapidly iterate through different hyperparameter configurations;
- Test multiple architectural variations;
- Optimize training strategies;
- Explore various loss function weightings.

Only after determining promising configurations on this smaller subset did it proceed to train the full model on the complete training set of 20,000 images. This two-phase approach significantly reduced the computational resources required for experimentation while still ensuring that the final model was trained on the full dataset for optimal performance.

The key hyperparameters optimized in the experiments included:

- Learning rate: 2×10^{-4} for both generator and discriminator;
- Batch size: 16;
- Loss function weights: $\lambda_{L1} = 100$ for the L1 loss component;
- Optimizer: Adam with $\beta_1 = 0.5$ and $\beta_2 = 0.999$;
- Dropout rate: 0.5 in the first three upsampling blocks of the generator.

2.3. Preprocessing

Each image underwent a preprocessing before training:

1. **Resizing:** All images were resized to 256×256 pixels to maintain consistent spatial dimensions while preserving sufficient facial detail for the colorization task.
2. **Color Space Conversion:** RGB images were converted to the CIELAB color space, which explicitly separates luminance (L) from chromaticity (a,b) channels. This transformation is particularly suitable for colorization tasks as it directly maps to the problem formulation (predicting a,b from L).
3. **Channel Normalization:**
 - The L channel was normalized to the range [0,1] by dividing by 100
 - The a,b channels were normalized to the range [-1,1] by dividing by 128
4. **Cropping:** During training, images were center-cropped to ensure consistent framing of facial features. For validation and testing, only center cropping was applied to maintain deterministic evaluation.

¹<https://www.kaggle.com/datasets/anitaascheri/celeba-short-dataset-training-validation-test>

2.4. Data Augmentation

More aggressive augmentation techniques—such as horizontal flipping, color jittering, or random affine transformations—were deliberately avoided in order to preserve natural facial geometry and chromatic characteristics. These properties are critical for the colorization task, where even slight distortions can lead to perceptually implausible or semantically inconsistent outputs.

3. Methods

For this work, Pix2Pix was selected over other GAN variants (e.g., CycleGAN or standard cGAN) due to its effectiveness in paired image translation tasks and its ability to preserve spatial detail through U-Net skip connections. An open-source Pix2Pix implementation² was used only as an initial reference. The final model diverges significantly, incorporating a revised generator/discriminator architecture, a modified training loop, and tailored evaluation metrics for facial image colorization.

3.1. Model Architecture

The model is based on the Pix2Pix architecture by Isola et al. [1], optimized for the colorization of grayscale facial images and it is an image-to-image translation architecture based on conditional generative adversarial networks (cGANs), employing a generative network and a discriminative network in an adversarial learning context.

Specifically, the generator takes as input the grayscale image (the L channel of the CIELAB color space) and outputs the two chromatic channels A and B, thereby reconstructing the colorized image. To preserve the structural details of the face, the generator follows a U-Net architecture, which combines low- and high-level information through skip connections. The discriminator, based on a PatchGAN network, receives as input both the grayscale image concatenated with either the real or generated color image, and evaluates whether the colorization proposed by the generator is plausible at a local level (on image patches). In this way, the discriminator guides the generator to produce more realistic and consistent colorizations.

The general model structure is illustrated in Figure 1.

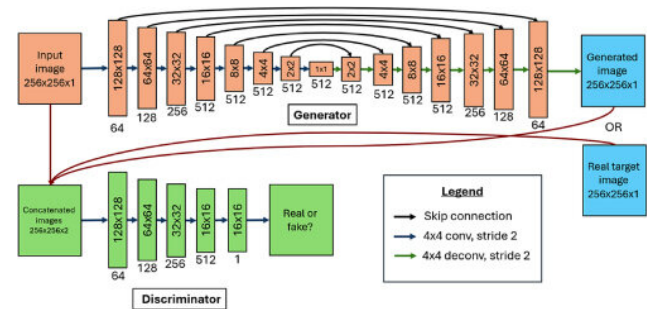


Figure 1. Pix2Pix network architecture displaying the U-Net generator and the PatchGAN discriminator [6].

3.1.1 Generator (U-Net)

The generator implements a U-Net architecture [7], an encoder-decoder structure with symmetric skip connections that link corresponding layers of encoder and decoder, facilitating the preservation of spatial details.

- **Encoder:** 7 downsampling blocks with 4×4 convolutions, stride 2, and padding 1; LeakyReLU activation (slope 0.2); batch normalization applied to all layers except the first; progressive channel expansion from 1 (L channel) up to 512.
- **Bottleneck:** a 2×2 convolutional layer with 512 feature maps, without batch normalization.
- **Decoder:** 7 upsampling blocks with 4×4 transposed convolutions, stride 2; ReLU activation; dropout ($p = 0.5$) applied to the first three layers; skip connections with encoder layers; final output layer uses a transposed convolution with tanh activation to generate normalized AB channels in $[-1, 1]$.

3.1.2 Discriminator (PatchGAN)

The discriminator adopts the PatchGAN architecture [1], which evaluates realism on 70×70 patches instead of the entire image, better capturing local textures and chromatic details.

- Input consists of concatenated L and AB channels (3 channels total);
- Four convolutional blocks with 4×4 filters, stride 2, and LeakyReLU activation (0.2);
- Channel progression: $3 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$;
- Batch normalization applied in all layers except the first;
- Final 1×1 convolution outputs a patch-wise probability map.

²<https://github.com/mberkay0/image-colorization>

3.2. Loss Function

The model optimization relies on a composite loss function that guides the generator during training by combining adversarial learning with direct supervision from ground truth data:

$$\mathcal{L}_G = \mathcal{L}_{\text{GAN}} + \lambda \cdot \mathcal{L}_{L1}$$

- **Adversarial loss ($\mathcal{L}_{\text{GAN}}$):** This term encourages the generator to produce colorized outputs that are indistinguishable from real images by a discriminator. It is based on binary cross-entropy and follows the standard GAN formulation [8]. The loss is computed as:

$$\mathcal{L}_{\text{GAN}} = \mathbb{E}[\log D(L, AB_{\text{real}})] + \mathbb{E}[\log(1 - D(L, G(L)))]$$

where D is the discriminator, L represents the input grayscale (L channel), AB_{real} are the ground truth chrominance channels, and $G(L)$ is the generator's prediction. The first term rewards the discriminator for correctly identifying real color images, while the second penalizes it for being fooled by the generator, thereby pushing G to produce more realistic colorizations.

- **Reconstruction loss ($\mathcal{L}_{L1}$):** This term measures the pixel-wise L1 distance (mean absolute error) between the predicted and ground truth AB channels:

$$\mathcal{L}_{L1} = \|AB_{\text{real}} - G(L)\|_1$$

Unlike the adversarial loss, which promotes perceptual realism, the L1 loss ensures color accuracy by directly supervising the generator's outputs. This helps preserve the underlying structure and chromatic fidelity of the original image, reducing artifacts and mode collapse.

- The scalar weighting factor λ balances the two loss components and is empirically set to $\lambda = 100$. This value is adopted from prior work (e.g., Pix2Pix [1]) and is chosen to ensure that the generator prioritizes producing images with realistic color fidelity while still maintaining visual plausibility.

3.3. Training Strategy and Hyperparameter Tuning

Training alternates updates between the generator and discriminator on mini-batches of size 16. Optimization is performed using the Adam optimizer (learning rate 0.0002, $\beta_1 = 0.5$, $\beta_2 = 0.999$), following established GAN practices [8]. Hyperparameters, including learning rate, batch size, and loss weights, were initially set according to prior work [1] and subsequently refined through empirical validation to maximize training stability and output quality.

To address common challenges in GAN-based colorization—such as adversarial instability, mode

collapse, and unrealistic chromatic saturation—a suite of stabilization techniques was employed:

- **Label smoothing:** Real labels are set to 0.9 to stabilize adversarial training and reduce discriminator overconfidence.
- **Random label flipping:** A small proportion (5%) of labels are randomly flipped during training to introduce regularization and prevent generator collapse [9].
- **Dropout:** Applied in early decoder layers to improve generalization and prevent overfitting.
- **Discriminator update frequency:** The discriminator is updated every two iterations to balance adversarial dynamics.
- **Gradient accumulation:** Simulates larger batch sizes under memory constraints.
- **Learning rate decay:** A linear decay is applied from epoch 35 to 50, reducing the rate by a factor of four to stabilize convergence.

3.4. Evaluation Metrics

Model performance is assessed on a held-out test set, completely separate from training and validation data to ensure unbiased generalization. A combination of quantitative and perceptual metrics is reported:

- **PSNR** (Peak Signal-to-Noise Ratio): pixel-wise fidelity;
- **SSIM** (Structural Similarity Index): perceptual and structural similarity;
- ΔE (**CIE76**): perceptual color difference in LAB space;
- **Chroma Accuracy:** a custom metric assessing saturation consistency in the A and B channels.

Generated LAB images are converted to RGB using standard color space transformations for comprehensive visual and chromatic quality assessment (Figure 2).

During training, a separate validation set was used to monitor progress and prevent overfitting, with metrics (PSNR, SSIM, ΔE , Chroma Accuracy) tracked periodically.



Figure 2. Example of input-output pipeline: input grayscale image (L channel), ground-truth color image (target LAB), and generated color output (LAB prediction converted to RGB).

3.5. Challenges and Solutions

The aforementioned stabilization strategies proved effective in overcoming the main difficulties of GAN-based colorization: adversarial instability, mode collapse, unrealistic color saturation, and out-of-gamut colors during LAB–RGB conversion. As a result, the final model is able to generate realistic, perceptually coherent, and chromatically accurate colorizations of grayscale facial images.

4. Experiments

The training strategy was organized into two main phases. First, an exploratory phase using a 2,000 image subset of the training data enabled rapid evaluation of different hyperparameter settings, minimizing computational requirements while efficiently identifying unstable configurations and convergence issues. This was followed by large-scale training on the complete set of 20,000 images, incorporating the best strategies identified earlier.

4.1. Preliminary Experiments

Four configurations were tested on the 2,000 image subset to assess the impact of learning rates, batch sizes, and regularization on GAN training for face colorization:

- **1st:** Equal learning rates (0.0002) for both generator and discriminator resulted in instability, with the discriminator quickly overpowering the generator (Figure 3).
- **2nd:** Lowering the discriminator’s learning rate to 0.001 and applying label smoothing improved stability, but hue shifts and oversaturation persisted (Figure 4).
- **3rd:** A smaller batch size and learning rate decay further stabilized training, yet the outputs appeared muted and less detailed (Figure 5).

- **4th:** Combining all the most effective strategies (adjusted learning rates, label smoothing, reduced batch size, and learning rate decay) provided the most stable training and improved image quality, with only minor color issues (Figure 6).



Figure 3. Training output at epoch 50, first configuration: poor color fidelity and structure.



Figure 4. Training output at epoch 50, second configuration: hue shifts, oversaturation.

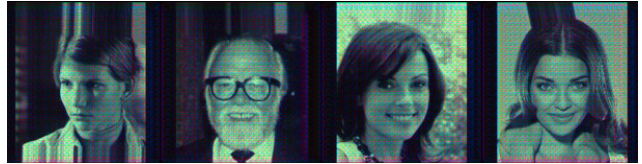


Figure 5. Training output at epoch 50, third configuration: muted tones, limited detail.



Figure 6. Training output at epoch 50, fourth configuration: sharper details, minor color issues.

4.2. Large-Scale Training and Optimization

Based on the fourth configuration, large-scale experiments were conducted on the full training set of 20,000 images. Two learning rate strategies were compared: one with a slower discriminator learning rate (0.0001) versus 0.0002 for the generator, and one using identical learning rates (0.0002 for both). The latter yielded slightly better results in terms of PSNR, SSIM, DeltaE, and Chroma Accuracy, and produced visually more coherent and stable outputs.

As a result, the configuration using equal learning rates for both generator and discriminator was adopted for final training. The model was trained for 100 epochs, with learning rate decay starting at epoch 60. Signs of overfitting appeared around epoch 60, as both quantitative metrics and visual quality on the validation set began to plateau.

4.3. Final Configuration

Insights gained from the previous large-scale experiments motivated further optimization of the training regime to improve efficiency and minimize overfitting. Specifically, the number of training epochs was reduced to 50, with the learning rate decay starting earlier at epoch 35. This adjustment led to comparable quantitative and qualitative results in substantially less time and with a reduced risk of overfitting.

4.3.1 Experimental Setup

The proposed model was trained for 50 epochs. Performance was monitored throughout training using quantitative metrics (PSNR, SSIM, DeltaE, Chroma Accuracy) in both the training and validation sets, and final results were computed on a held-out test set. Loss curves and evaluation metrics were saved and visualized for further analysis.

The main training hyperparameters were as follows:

- **Epochs:** 50 (total training epochs)
- **Batch size:** 16 (optimized for memory constraints)
- **Image resolution:** 256×256 pixels
- **Optimizer:** Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$)
- **Initial learning rate:** 2×10^{-4} (for both generator and discriminator)
- **Weight decay:** 1×10^{-4} (L2 regularization)
- **Loss weight:** $\lambda = 100$ (for L1 reconstruction loss)

4.3.2 Training Curves

During training, the generator and discriminator losses were monitored to assess convergence and training stability (Figure 7). At the beginning of the training process, the generator loss exhibits a sharp peak, followed by a rapid decrease as the model starts to learn. This initial peak is typically observed in GAN training and is due to the generator’s lack of knowledge at the outset, resulting in outputs that are easily distinguished by the discriminator and, therefore, a high initial loss. As training progresses, the generator rapidly improves, leading to a sharp reduction in loss. Subsequently, the generator loss progressively declines and stabilizes, with minor oscillations, indicating effective learning

and adaptation. The discriminator loss, in contrast, starts at a moderate value and quickly reaches a steady regime, maintaining relatively low and stable values throughout the remainder of training. The overall trend demonstrates that both networks reach a dynamic equilibrium without signs of divergence or instability, suggesting that the adopted training strategies (such as label smoothing, label flipping, and learning rate decay) contributed to a robust and stable adversarial training process.

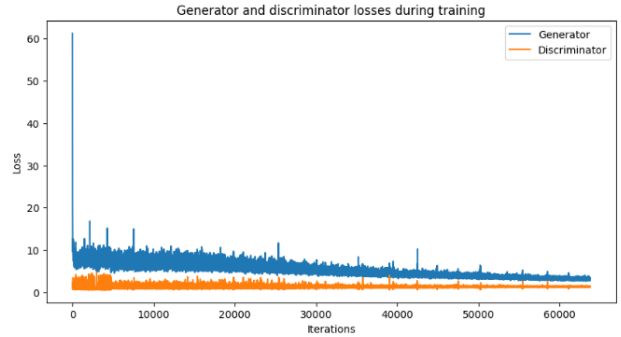


Figure 7. Generator and discriminator loss curves across epochs.

4.3.3 Quantitative Results

Table 1 summarizes the main quantitative results obtained on the test set, averaged over all samples.

Table 1. Quantitative Results on the Test Set

Metric	Value
PSNR	26.1813
SSIM	0.9455
DeltaE	10.7202
Chroma Accuracy	0.5992

To quantitatively evaluate the performance of the proposed model, several standard image quality metrics were computed on the test set: PSNR, SSIM, DeltaE, and Chroma Accuracy.

The *Peak Signal-to-Noise Ratio (PSNR)*, expressed in decibels, measures the fidelity of the reconstructed images with respect to the ground truth; higher values indicate lower levels of distortion. The proposed model achieved a PSNR of **26.18 dB**, which is considered a good result for automatic colorization tasks.

The *Structural Similarity Index Measure (SSIM)* is a dimensionless metric ranging from 0 to 1 that assesses the perceptual similarity between generated and reference images by considering luminance, contrast, and structural information. The obtained value of **0.9455** indicates that the generated images are highly similar to the originals from a perceptual standpoint.

The *DeltaE* metric, also dimensionless, quantifies the perceived color difference between generated and ground-

truth images; lower values correspond to more accurate color reproduction. The value of **10.72** reflects a moderate yet acceptable chromatic difference for this type of task, where perfect color matching is not strictly required.

Finally, *Chroma Accuracy* is a dimensionless metric ranging from 0 to 1 that measures the model’s ability to accurately reconstruct chrominance information. The model achieved a Chroma Accuracy of **0.599**, indicating good consistency in color saturation across the entire test set.

Overall, these results confirm that the proposed approach is capable of generating visually compelling and chromatically faithful colorized images.

4.3.4 Qualitative Results

Figure 8 displays qualitative examples from the test set. Each row shows, from left to right: the input grayscale image (L channel), the ground-truth color image, and the output generated by the model. The results confirm that the system is capable of recovering realistic and visually plausible colorizations, preserving both global facial structure and local details.



Figure 8. Qualitative results on test images: input grayscale, ground-truth, and generated colorization.

4.3.5 Comparison and Ablation Analysis

Compared to the main works in the literature on automatic image colorization, this model is distinguished by the explicit integration of several training stabilization techniques, such as label smoothing, label flipping, and learning rate decay. Although these strategies have been widely studied for stabilizing GAN training in other contexts [9, 10], they

are rarely employed in colorization research, where the focus has traditionally been on architectural or perceptual aspects [1, 2] rather than on training regularization strategies.

The ablation study conducted demonstrates that these techniques significantly contribute both to the stability of the training process and to the visual quality of the results obtained. Specifically, three key aspects were analyzed: label smoothing, label flipping, and learning rate decay. For each technique, an ablation experiment was performed in which the corresponding component was removed or disabled, while all other hyperparameters and the model architecture were kept unchanged; the results were then compared with those of the full system.

- **No Label Smoothing:** Removing label smoothing led to increased instability and erratic loss curves; the final PSNR dropped by ~ 0.5 dB.
- **No Label Flipping:** Disabling random label flipping caused faster discriminator convergence but increased the risk of mode collapse and color artifacts.
- **No Learning Rate Decay:** Training without progressive learning rate decay led to earlier plateauing of validation metrics and a degradation of 0.2–0.3 absolute points in PSNR/SSIM.

Due to differences in dataset splits, preprocessing, and evaluation protocols, a direct quantitative comparison with prior work is not possible. Nevertheless, the results obtained are in line with, or superior to, typical values reported in the literature for GAN-based face colorization.

5. Conclusions

This work presented an optimized implementation of the Pix2Pix architecture, tailored for the automatic colorization of grayscale facial images. By systematically exploring hyperparameters and stabilization strategies, this study demonstrated that a combination of label smoothing, random label flipping, and progressive learning rate decay can substantially improve the stability and visual fidelity of GAN-based colorization models. Quantitative results on the reduced CelebA dataset confirmed that the proposed approach achieves strong performance across multiple metrics, with realistic and perceptually convincing outputs that are comparable to or exceed established baselines.

Beyond the final performance metrics, the experimental process provided important insights into the dynamics of GAN training for colorization. It was observed that careful balance between generator and discriminator learning rates, as well as regularization through label manipulation and learning rate scheduling, are critical to avoiding mode collapse and ensuring consistent convergence. The

two-phase training regime—beginning with rapid prototyping on a reduced dataset and followed by full-scale optimization—proved effective in reducing computational costs while enabling robust hyperparameter selection.

Despite these successes, several challenges remain. The model occasionally generates less vibrant or slightly inaccurate colors, particularly for rare facial attributes or lighting conditions underrepresented in the training set. Furthermore, the reliance on paired data (grayscale/color) limits applicability to domains where such data is not readily available.

Future work could focus on integrating semantic segmentation to further guide the colorization process, as well as exploring self-supervised or unpaired learning approaches (e.g., CycleGAN) to relax the need for paired datasets. Additional research into perceptual loss functions and adaptive regularization strategies might also further enhance the realism and diversity of generated colorizations. Overall, the findings highlight the effectiveness of the Pix2Pix architecture for face colorization and provide a solid foundation for further advances in this area.

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